

Dual-Uncertainty modeling in financial time-series via VMD-LSTM with concrete dropout and VMD-WGAN

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Financial time-series forecasting requires not only accurate point predictions but also a principled characterization of the uncertainty around future outcomes. We proposed a unified dual-path framework that modeled both forms using a shared variational mode decomposition (VMD) backbone. VMD stabilized nonstationary signals, enabling the predictive path—a long short-term memory (LSTM) network integrated with concrete dropout—to quantify epistemic uncertainty, while the generative path used a conditional Wasserstein generative adversarial network (WGAN) to capture aleatoric risk. Empirical evaluations on the Standard & Poor's 500 (S&P 500) index and Financial Times Stock Exchange 100 (FTSE 100) index demonstrated superior predictive accuracy and distributional fidelity over strong baselines. Comprehensive ablation studies and regime-conditioned analyses revealed a clear frequency-wise division of labor: Low-frequency modes drove predictive accuracy, while the generative path successfully reproduced heavy-tailed, regime-dependent return distributions. These findings underscored the efficacy of decomposed uncertainty modeling for robust financial risk assessment.

Keywords: stock prediction, variational mode decomposition, deep learning, generative adversarial networks, uncertainty modeling

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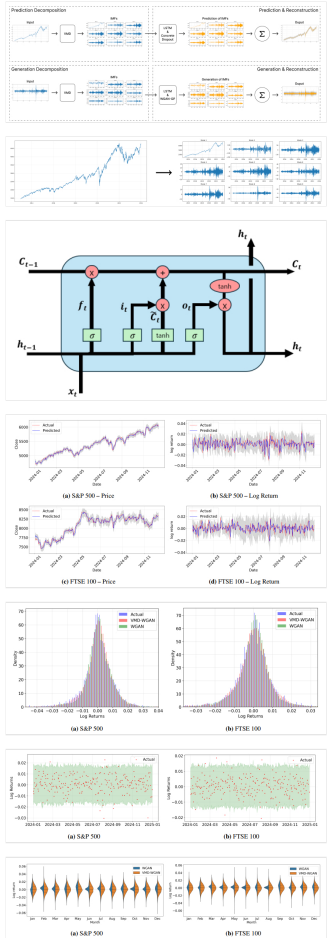
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